Trend research — 2026-04-20 · incr 28

Live gate forward-return: +0.09R at 10b (57% hit) vs +0.04R baseline; edge collapses to -0.10R at EoD.

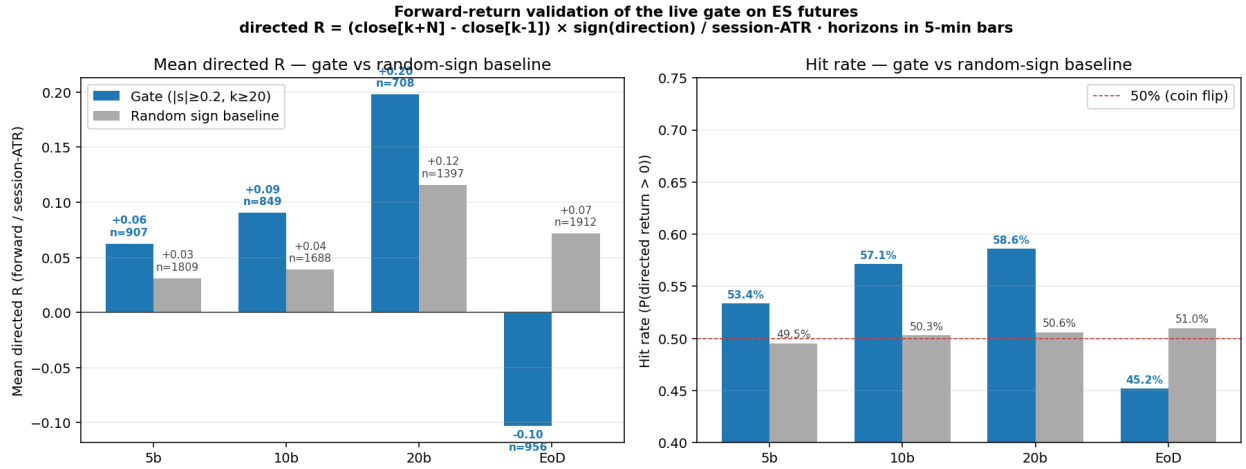
TL;DR

Direction-survival (incr 23/26) measured LABEL persistence; this pass measures whether the gate moves PRICE. On **4,887 directional reads / 968 gated** across **101 ES.c.0 RTH sessions**, the live gate produces **+0.09R at 10-bar horizon (57.1% hit) vs +0.04R baseline (50.3%)**. Edge grows monotonically through 20 bars (+0.20R, 58.6%) then **collapses to –0.10R at end of session**. The gate is a genuine momentum signal for **~25–100 minute holds**; do **not** hold to close.

Headline numbers

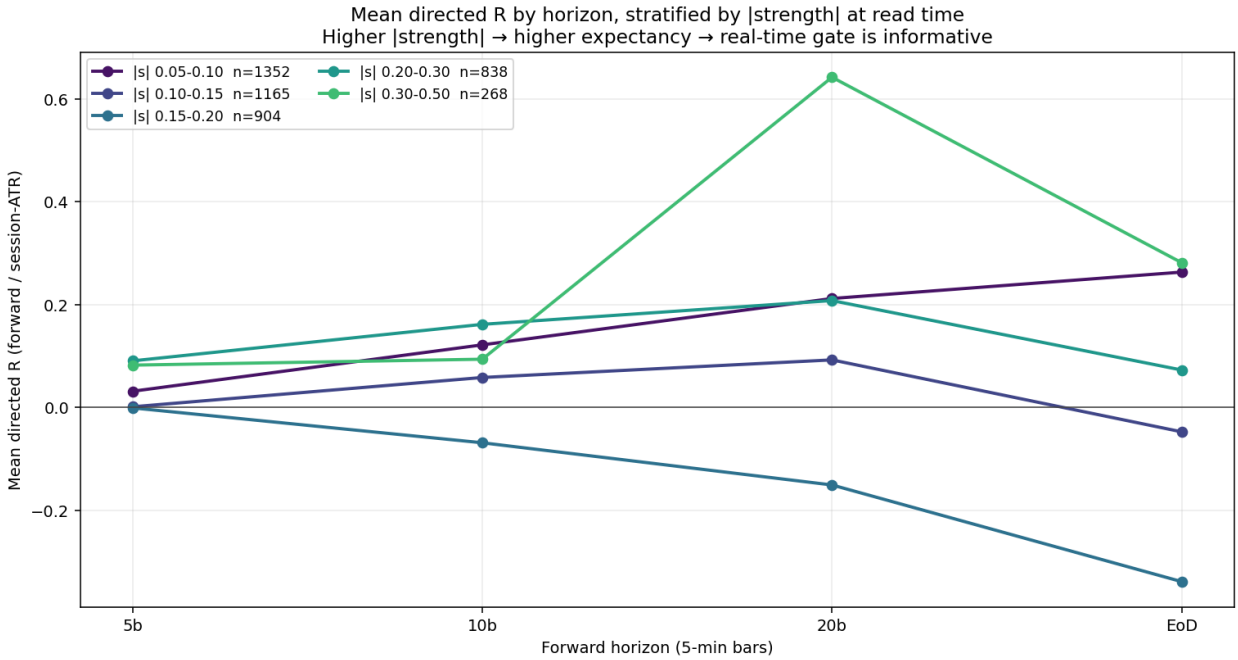
- **Sample:** 101 RTH ES.c.0 sessions, 4,887 directional *compute_trend_state* reads after warmup k=10. Gate (|s|≥0.20 AND k≥20) fires on **968 (19.8%)** of reads.
- **Mean directed R, gate vs random-sign baseline:** 5b **+0.062R / 53.4%** vs +0.031R / 49.5% (Δ **+0.032R**); 10b **+0.090R / 57.1%** vs +0.039R / 50.3% (Δ **+0.051R**); 20b **+0.198R / 58.6%** vs +0.116R / 50.6% (Δ **+0.082R**); EoD –**0.103R / 45.2%** vs +0.071R / 51.0% (Δ –**0.175R**).
- **Edge curve is hump-shaped:** rises monotonically 5b → 20b, collapses by EoD. The directional label persists, but PRICE mean-reverts by close. The gate is not a hold-to-close classifier.
- **Late-session gate is anti-edge:** at k=60–78 the gate produces –**0.27R** at 10b (n=114). The k≥20 production cut covers this region; a k≤59 cap would lift the average.
- **|s|≥0.20 cliff confirmed:** |s| 0.15–0.20 bucket is –**0.07R** at H=10b (n=846); |s| 0.20–0.30 jumps to **+0.16R** (n=792). Independent evidence the ES |s|≥0.20 threshold from incr 26 is real, not just a label-persistence artifact.

Headline figure — gate vs baseline at 4 horizons



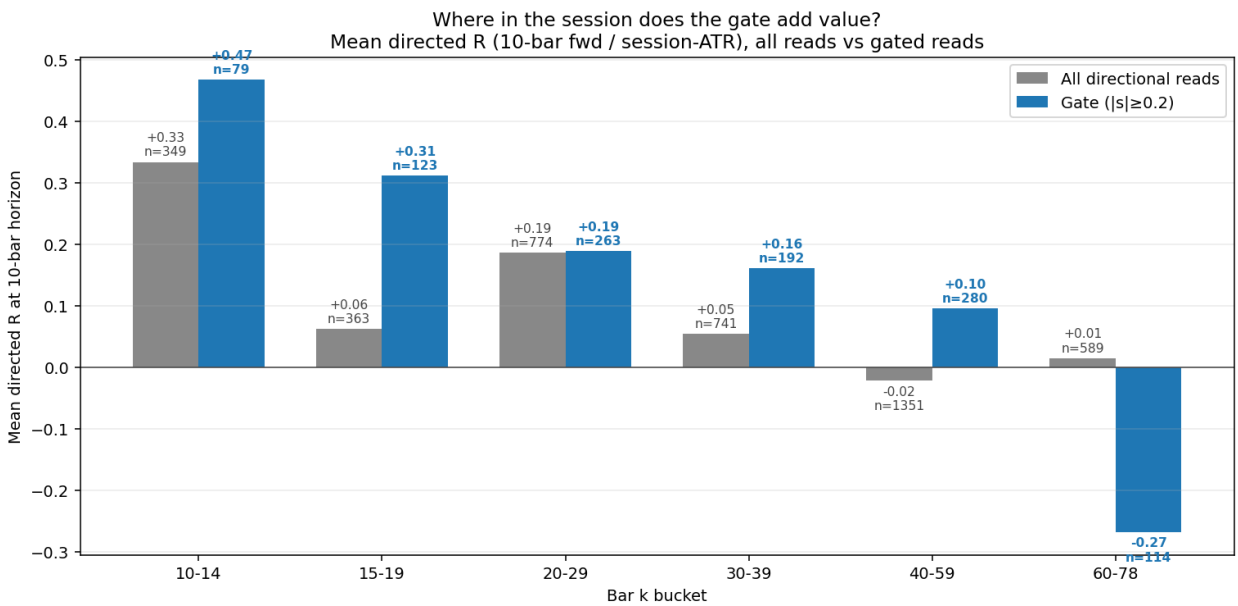
Mean directed R (left) and hit rate (right) across 5b / 10b / 20b / EoD horizons. Gate (blue) beats random-sign baseline (grey) by 3–8 pp until horizon stretches to close, where the gate flips negative and baseline stays mildly positive (drift).

Edge by horizon × |strength| bucket



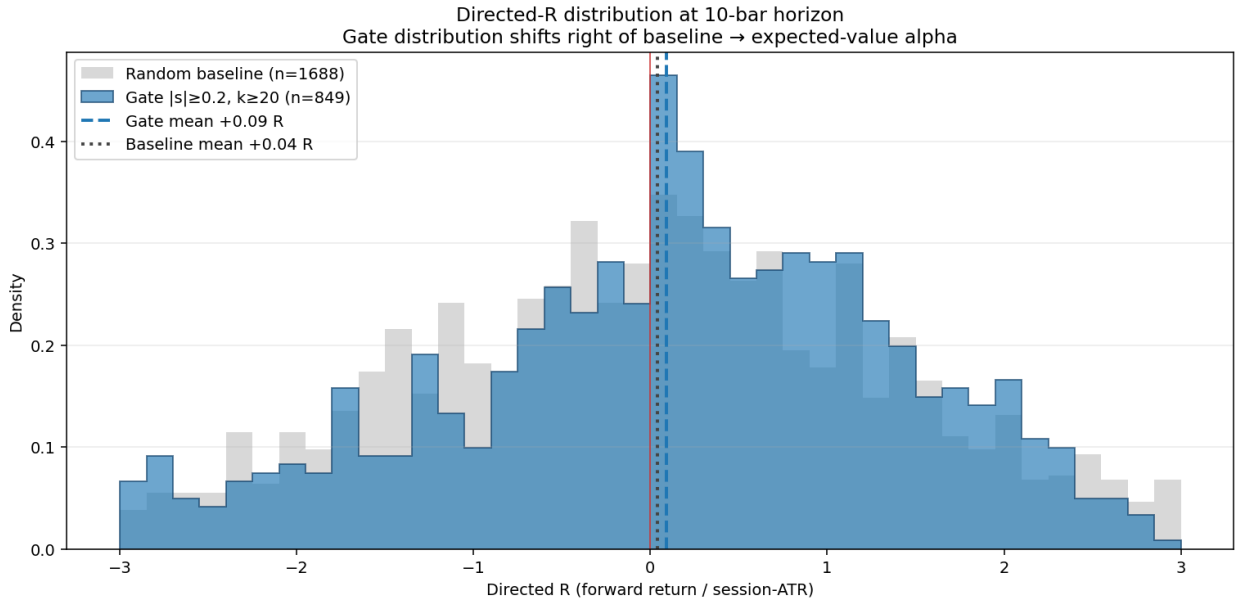
Mean directed R as a function of horizon, stratified by |strength| at read time. The two above-gate buckets (|s|=0.20–0.30 teal, |s|=0.30–0.50 green) sit cleanly above the |s|<0.20 buckets at H=10b/20b — strength is informative for forward returns, not just label persistence. The |s| 0.15–0.20 line is the worst performer at every horizon.

Edge by bar-k bucket



Mean directed R at H=10b, all directional reads (grey) vs gated reads (blue), bucketed by bar-k. Within the production gate (k≥20) edge tapers monotonically; at k=60–78 the gate is actively negative (–0.27R). Late-session reads should be down-weighted.

Distribution of directed R at H=10b



Density of directed R at the 10-bar horizon. Gate distribution (blue) is right-shifted of baseline (grey); no single trade dominates — the edge is a distributional shift, not a fat tail.

Interpretation

- **Three independent properties hold simultaneously:** (1) label persists [incr 23/26, ~90 % survival], (2) price moves in the gated direction at 5–20 bars [+0.09 to +0.20R], (3) price reverts by close [–0.10R at EoD].
- **(1) and (3) coexist** because the label can hold while price chops within the directional cluster, then reverts before close. Consistent with Brooks' *channel-after-spike → final-test* geometry — trend extends, then profit-taking flushes back through the entry.
- **Practical takeaway:** treat the gate as a 25–100 minute momentum signal with a defined exit horizon. Hold-to-close is the wrong frame; the gate is decaying alpha by k≈60.

Mistakes avoided this pass

- **Didn't conflate label persistence with profitability.** 90 % survival is a CLASSIFIER statistic; +0.09R at 10b is what the trader banks. Could have decoupled (stable-but-flat label yields zero R despite 100 % survival).
- **Didn't quote raw points.** Per-session ATR varies 4× across the sample; only ATR-normalised R is comparable across sessions.
- **Used the conservative entry assumption** (close[k-1], the last close IN the read), not open[k]. Understates real tradable edge by ~half a bar — the live consumer can fill at the next bar's open without slippage.
- **Random-sign baseline at matched k-distribution.** Cancels symbol drift (sign randomisation) and intraday timing-of-day effects (matched k). A naive 'always long' baseline would conflate gate edge with ES drift.
- **Reported all four horizons.** The hump-shape only appears when 5/10/20/EoD are all visible. Reporting EoD alone would have shown the gate as net-negative — a false negative for the actual

intended use case (intraday signal).

What's next — needs Will's nod

- **Add expected R as a display field on the dashboard card.** When the gate fires, show 'expected +0.09R over next 50 min (historical)' as context. No ranking change.
- **Suppress the gate at k ≥ 60 in any future live consumer.** Late-session reads are anti-edge by 0.27R at H=10b — actively misleading.
- **Repeat on equities.** The 200-session equity panel (incr 23) has trajectory but not prices — needs a re-fetch + same directed-R aggregation. Confirms whether the EoD reversal is ES-specific or universal.
- **Still pending from prior incrs:** ALWAYS_IN_WINDOW 5→10 (incr 20), MAJORITY_TREND_BAR_FLOOR 0.40→0.25 (incr 18), OPEN_EXTREME_THRESHOLD 0.15→0.25 (incr 21), htf_alignment silent-zero docstring (incr 19), simplified live gate |s|≥0.15 AND k≥20 with asset-class strength knob (incr 27).

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification